BRAINSTEM & CEREBELLUM

ANNOTATION GUIDELINES

Annotation Review & Definition Refinement Framework

Radiology Image Annotation & Quality Validation Tool | June 2026

Prepared by	Zainab Malik
Institution	Health Informatics, Deggendorf Institute of Technology
Project	Radiology Image Annotation & Quality Validation Tool
Scope	Posterior Fossa (Brainstem & Cerebellum) + Chest X-Ray
Imaging Modalities	MRI (T1, T2, FLAIR, DWI/ADC, T1+Gad) and CT
Tool Stack	Python 3.11, Streamlit, pandas, Pillow, pydicom, numpy
GitHub	github.com/zainabmalik1532004-ux/radiology-annotation-validation-tool

1. Introduction & Purpose

This document provides the annotation guidelines, quality control criteria, and validation framework for the Radiology Image Annotation & Quality Validation Tool. It covers annotation review and annotation definition refinement workflows for brainstem, cerebellum, and chest X-ray imaging datasets used in AI model training.

Accurate annotation of the brainstem and cerebellum is critical for training AI models to detect neurological conditions such as posterior fossa tumors, multiple sclerosis plaques, cerebellar atrophy, and brainstem infarctions.

1.1 Scope of This Document

- Anatomical definitions for all key brainstem and cerebellar structures
- Annotation rules for each structure on MRI (T1, T2, FLAIR) and CT
- Quality control criteria for annotation review
- Common annotation errors and how to avoid them
- Validation checklist before dataset handoff to AI pipeline
- Tool field reference, how each annotation field maps to the guidelines.

1.2 Target Users

- Medical annotators reviewing and labeling neuroimaging datasets
- Annotation team leads defining and refining labeling standards

- AI researchers validating training data quality
- R&D teams building radiology AI models.

2. Project Overview, Annotation & Validation Tool

The tool is a Python/Streamlit web application that operationalizes these annotation guidelines. It enforces annotation review rules through mandatory form fields, and supports annotation definition refinement by providing a structured, versioned framework for labeling decisions.

Step	What Happens in the Tool	Guideline Section
Upload Image	Annotator uploads chest X-ray or brain MRI/CT	Section 4 — Imaging Modalities
Select Scan Type	Choose Chest X-Ray or Brainstem & Cerebellum mode	Sections 5.2 & 5.3 — Structure Rules
Select Structure	Pick anatomical structure from guideline-defined dropdown	Sections 5.2 & 5.3 — Structure Definitions
Select Imaging Sequence	Record T1/T2/FLAIR/DWI/CT — mandatory field	Section 4 — Sequence Reference Rule
Select Laterality	Left/Right for bilateral structures — enforced	Section 5.1 — Bilateral Labeling Rule
Boundary Certainty Flag	Mark certain or uncertain boundary	Section 6.1 — Boundary Flags Check
Clinical Flag	Flag Chiari, artifact, asymmetry, demyelination	Sections 5.3.3 & 6.1 — Clinical Flags
Confidence & Notes	Rate certainty 1-10; document observations	Section 6.1 — Quality Threshold
Save & Validate	Annotation saved; dashboard runs quality checks	Sections 6.1 & 6.2 — Annotation Review

3. Anatomical Overview

Before performing any annotation, the annotator must have a clear understanding of the structures involved. Every structure in the tool dropdown corresponds to a precisely defined anatomical region described in this section.

3.1 The Brainstem

Structure	Location	Key Function	Clinical Relevance
Midbrain	Superior brainstem, below thalamus	Visual & auditory reflexes, motor control	Parinaud syndrome, Weber syndrome
Pons	Middle brainstem, anterior to cerebellum	Breathing regulation, facial sensation	Central pontine myelinolysis, MS plaques
Medulla Oblongata	Inferior brainstem, transitions to spinal cord	Heart rate, breathing, swallowing	Lateral medullary syndrome (Wallenberg)

3.1.1 Midbrain Landmarks

- Cerebral aqueduct (Aqueduct of Sylvius) midline CSF channel
- Superior and inferior colliculi, visible as paired bumps on dorsal surface
- Cerebral peduncles paired white matter tracts on ventral surface.
- Red nucleus, round structure visible on T2 as hypointense

- Substantia nigra, crescent-shaped, hypointense on T2

3.1.2 Pons Landmarks

- Basilar artery impression, midline groove on anterior surface
- Fourth ventricle floor (pontine tegmentum)
- Pontine nuclei scattered gray matter within white matter.
- Transverse pontine fibers, horizontal white matter bands
- Middle cerebellar peduncles, connecting pons to cerebellum.

3.1.3 Medulla Oblongata Landmarks

- Pyramids paired anterior ridges containing corticospinal tracts.
- Olives, oval prominences lateral to pyramids.
- Obex, marks transition to spinal cord / caudal 4th ventricle
- Inferior cerebellar peduncles, posterior lateral connections to cerebellum

3.2 The Cerebellum

Region	Description	Function	Imaging Appearance
Cerebellar Vermis	Central midline, connects hemispheres	Axial balance, gait coordination	T1: isointense, tree-like folial pattern
Cerebellar Hemispheres	Two lateral lobes (Left/Right)	Limb coordination, fine motor skills	T1/T2: foliated gray matter cortex
Cerebellar Tonsils	Inferior lobules near foramen magnum	Important landmark	Critical: position relative to foramen magnum
Dentate Nucleus	Deep cerebellar nucleus, lateral WM	Relay motor signals to thalamus	T2: hypo intense due to iron. depositions
Flocculonodular Lobe	Anterior inferior cerebellum	Vestibular function, eye movement	Small, adjacent to 4th ventricle

3.2.1 Cerebellar Peduncles

Peduncle	Connection	Imaging Level	Main Tract
Superior Cerebellar Peduncle (SCP)	Cerebellum to Midbrain	Midbrain/pons junction	Dentato-rubro-thalamic tract
Middle Cerebellar Peduncle (MCP)	Cerebellum to Pons	Mid-pons level	Pontocerebellar fibers
Inferior Cerebellar Peduncle (ICP)	Cerebellum to Medulla	Medullary level	Spinocerebellar tracts

4. Imaging Modalities & Signal Characteristics

The tool requires annotators to select the imaging sequence for every annotation. Selecting the wrong sequence is one of the most common annotation errors.

4.1 MRI Sequences Posterior Fossa

Sequence	Brainstem Appearance	Cerebellum Appearance	Best Used For
T1-weighted	Isointense gray/white matter contrast	Clear folial pattern, white matter bright	Anatomy, boundaries, atrophy
T2-weighted	CSF bright; nuclei visible by iron contrast	Gray matter bright; dentate nucleus dark	Edema, demyelination, infarction

FLAIR	CSF suppressed; parenchymal lesions bright	Lesions in cortex/white matter highlighted	MS plaques, ischemic changes
DWI/ADC	Acute infarcts bright on DWI/dark on ADC	Acute cerebellar infarcts visible	Acute stroke detection
T1 + Gadolinium	Enhancement = blood-brain barrier breakdown	Tumor enhancement, metastases	Tumors, abscesses, inflammation

4.2 CT Posterior Fossa Considerations

- Normal brainstem: isodense to gray matter (30-40 HU)
- Acute hemorrhage: hyperdense (60-90 HU) annotate as hemorrhage, not normal tissue.
- Beam hardening artifact: dark bands between petrous bones DO NOT annotate as lesion.
- Cerebellar hemispheres: symmetric, foliated appearance asymmetry warrants flag.

5. Annotation Definition Rules

This section defines the precise rules for annotating each structure. These rules form the core of annotation definition refinement ensuring all team members apply identical standards. The tool enforces these rules through mandatory fields, structured dropdowns, and automated validation checks.

5.1 General Annotation Principles

- Use the minimum bounding region that fully includes the target structure
- Do not include adjacent CSF spaces within a parenchymal annotation
- When structure boundaries are unclear, use the 'Uncertain needs review' boundary flag.
- All annotations must reference the specific imaging sequence mandatory field in tool.
- Bilateral structures require separate Left/Right labels enforced by the tool's laterality field.
- Do not annotate artifacts use the 'Beam hardening artifact' clinical flag instead.

5.2 Brainstem Annotation Rules

5.2.1 Whole Brainstem Segmentation

Rule	Instruction
Superior boundary	Slice where cerebral aqueduct first becomes visible marks midbrain-diencephalon junction
Inferior boundary	Slice at the level of the foramen magnum where pyramidal decussation is visible
Lateral boundaries	Exclude the cerebellar peduncles unless specifically requested in task definition
Anterior boundary	Exclude the basilar artery and surrounding CSF
Posterior boundary	Include the 4th ventricle roof; exclude the cerebellum

5.2.2 Sub-structure: Midbrain

- Superior colliculi: annotate as paired oval structures on dorsal midbrain; T2 isointense
- Inferior colliculi: smaller, more inferior to superior colliculi; part of auditory pathway
- Cerebral peduncles: butterfly-shaped white matter on ventral midbrain; T1 hyperintense
- Cerebral aqueduct: do not annotate as tissue label as CSF space only.

5.2.3 Sub-structure: Pons

- Basis pontis: anterior bulge of pons; annotate separately from tegmentum when required
- Pontine tegmentum: dorsal pons including cranial nerve nuclei (CN V-VIII)
- Middle cerebellar peduncles: lateral white matter tracts include fully up to cerebellar white matter.
- 4th ventricle floor: posterior pons surface label as boundary, not parenchyma

5.2.4 Sub-structure: Medulla

- Pyramids: anterior paired structures; include full length from pons junction to decussation
- Inferior olivary nuclei: lateral to pyramids; visible as oval gray matter on T2
- Posterior columns: dorsal medulla; annotate separately only on high-resolution sequences

5.3 Cerebellum Annotation Rules

5.3.1 Cerebellar Hemisphere Segmentation

- Medial boundary: cerebellar vermis (label vermis separately)
- Superior boundary: tentorium cerebelli does not include supratentorial structures.
- Inferior boundary: cerebellar tonsils (include within hemisphere unless specifically isolating)
- Anterior boundary: cerebellar peduncles include up to peduncle attachment point.

5.3.2 Cerebellar Vermis

- Midline structure divide at the midline on axial and coronal views
- Superior vermis: above the primary fissure (lobules I-V)
- Inferior vermis: below primary fissure (lobules VI-X); includes nodulus and uvula
- Do not include the vallecula (CSF space between tonsils below vermis)

5.3.3 Cerebellar Tonsils Clinical Importance

CLINICAL FLAG: Cerebellar tonsils extending more than 5mm below the foramen magnum indicates possible Chiari malformation Type I. Select 'Tonsil descent > 5mm possible Chiari Type I' in the tool's clinical flag dropdown.

- Normal position: at or above the level of the foramen magnum
- Measurement required: distance from tip of tonsil to foramen magnum line (McRae's line)
- Annotation: mark tip of tonsil and reference line on sagittal T1

5.3.4 Dentate Nucleus

- Located within the white matter of each cerebellar hemisphere
- Appears as crinkled, tooth-like gray matter on high-resolution T2
- Characteristically hypointense on T2 due to iron deposition do not flag as pathology.
- Bilateral annotations required left and right labeled separately (enforced by laterality field)

6. Quality Control & Annotation Review Criteria

This section defines the criteria used during annotation review the process of checking whether completed annotations are medically accurate and technically correct. All checks in Section 6.1 are automated by the tool Validation Dashboard.

6.1 Annotation Review Checklist

Check	Criteria	Tool Implementation	Pass/Fail/Flag
Boundary accuracy	Boundary follows anatomical margins within 2mm	Boundary certainty field	Fail if > 2mm
Structure completeness	Entire target structure included	Structure dropdown from guideline list	Fail if part absent
CSF exclusion	No CSF spaces in solid tissue annotation	Notes + annotation review	Fail if CSF included
Artifact labeling	No artifacts annotated as pathology	Clinical flag: Beam hardening option	Flag for review
Bilateral symmetry	L/R structures labeled independently	Laterality field enforces L/R	Flag if asymmetry
Sequence reference	Annotation tied to correct sequence	Imaging sequence mandatory field	Fail if wrong
Boundary flags	Uncertain boundaries marked	Boundary certainty field	Pass only if flagged
Clinical flags	Abnormal findings flagged	Clinical flag dropdown in form	Fail if missed

6.2 Annotation Definition Refinement Process

1. Identify the systematic error through annotation review
2. Document the exact ambiguity in the original guideline
3. Propose a refined definition with explicit criteria
4. Validate with a radiologist or medical expert
5. Update this guideline document with version note
6. Re-annotate affected cases using the refined definition

6.3 Common Annotation Errors Posterior Fossa

Error Type	Description	Prevention
Tonsil-hemisphere confusion	Cerebellar tonsils included within hemisphere label	Always segment tonsils separately on sagittal view first
Peduncle exclusion	Cerebellar peduncles excluded from brainstem boundary	Follow boundary rules in Section 5.2.1 strictly
CT artifact as lesion	Beam hardening artifact labeled as infarction	Use clinical flag; always cross-reference MRI if available
Vermis-hemisphere overlap	Vermis boundary not clearly separated from hemispheres	Use midline sagittal slice to define vermis first
CSF inclusion in brainstem	Peribrainstem cisterns included in brainstem annotation	Exclude all CSF — annotate only parenchymal tissue
Wrong sequence	T1 annotation applied to T2 image or vice versa	Always record sequence type at start of session
Missed pathology flag	Abnormal finding annotated as normal without flag	Any asymmetry requires a flag — no exceptions

6.4 Automated Validation Dashboard

Quality Check	Threshold	Alert Type	Action Required

Low confidence annotation	Confidence < 5/10	Warning (yellow)	Re-annotate before handoff
Missing notes	Notes field empty	Warning (yellow)	Add documentation
Uncertain boundary flag	Boundary = Uncertain	Warning (yellow)	Escalate for radiologist review
Clinical flag present	Any flag other than None	Error (red)	Expert sign-off required
Inter-annotator agreement	Dice coefficient < 0.80	Pre-handoff check	Re-review and adjudicate

7. Pre-Handoff Validation Checklist

Before any annotated dataset is handed off to the AI pipeline, the following validation must be completed.

#	Validation Item	Tool / Manual	Status
1	All target structures annotated for 100% of cases in dataset	Manual review	[ ] Complete
2	Boundary accuracy verified against review checklist (Section 6.1)	Tool flags uncertain boundaries	[ ] Complete
3	Bilateral structures labeled independently (L/R separate)	Tool enforces laterality field	[ ] Complete
4	All imaging sequence references documented per annotation	Tool mandatory field	[ ] Complete
5	Clinical flags applied for all abnormal findings	Tool clinical flag field	[ ] Complete
6	All uncertain boundary flags reviewed and resolved or escalated	Tool dashboard warning	[ ] Complete
7	Inter-annotator agreement calculated and above threshold (>0.80 Dice)	Manual calculation	[ ] Complete
8	Artifact annotations removed from dataset	Tool clinical flag filter	[ ] Complete
9	Annotation definition version recorded in dataset metadata	Manual	[ ] Complete
10	Final review by medical expert completed and signed off	Manual	[ ] Complete

8. Glossary of Key Terms

Term	Definition
Annotation	The process of labeling structures or findings in a medical image for AI training
Annotation Definition Refinement	Updating labeling rules based on systematic review of errors or ambiguities
Annotation Review	Quality check of completed annotations against defined criteria
Arbor Vitae	The branching white matter of the cerebellum, visible on sagittal MRI as a tree-like structure
Beam Hardening	CT artifact causing dark bands in the posterior fossa due to X-ray absorption by bone
Cervicomedullary Junction	Anatomical transition point between medulla oblongata and spinal cord at foramen magnum
Chiari Malformation	Structural defect where cerebellar tonsils herniate below the foramen magnum
Dentate Nucleus	Largest deep cerebellar nucleus; appears hypointense on T2 due to iron content
Dice Coefficient	Statistical measure of annotation overlap between two annotators; range 0-1 (1 = perfect)
FLAIR	Fluid Attenuated Inversion Recovery — MRI sequence that suppresses CSF signal
Foramen Magnum	Large opening at base of skull through which brainstem transitions to spinal cord
Peduncle	White matter tract connecting cerebellum to brainstem (superior, middle, inferior)

Posterior Fossa	Inferior compartment of the skull containing brainstem and cerebellum
Segmentation	Delineating the full extent of an anatomical structure boundary in imaging
Tentorium Cerebelli	Dural fold separating cerebellum (infratentorial) from cerebrum (supratentorial)
Vermis	Central midline structure of the cerebellum connecting the two hemispheres